HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2018)

Student No.: 09 Name: _____ Marks: / 16

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. Let a , b and c be non-zero numbers. If $3a = 4b$ and $a : c = 2 : 5$, then $\dfrac{a+3b}{b+3c} =$

   A. $\dfrac{5}{3}$.

   B. $\dfrac{13}{33}$.

   C. $\dfrac{30}{53}$.

   D. $\dfrac{75}{38}$.

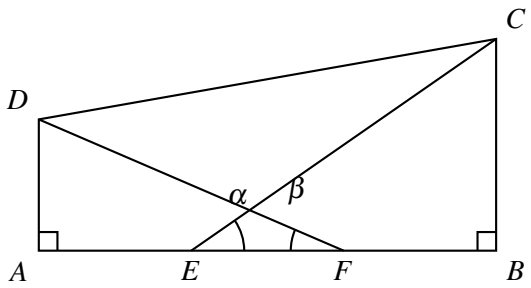
(DSE-2018 PII Q10)

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2. In the figure, $ABCD$ is a trapezium with $\angle ABC = \angle BAD = 90^\circ$. E and F are points lying on AB such that E and F divide AB into three equal parts. Which of the following must be true?



I. $AF \sin\alpha = BE \sin\beta$

II. $CE \cos\alpha = DF \cos\beta$

III. $AD \tan\alpha = BC \tan\beta$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2018 PII Q21)

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3. The polar coordinates of the points C , D and E are $(16, 127^\circ)$, $(12, 217^\circ)$ and $(5, 307^\circ)$ respectively. Find the perimeter of $\triangle CDE$.

A. 54

B. 78

C. 126

D. 130

(DSE-2018 PII Q24)

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4. Consider the following system of inequalities:

$$\begin{cases} x-21 \le 0 \\ x-y-35 \le 0 \\ x+5y-91 \le 0 \\ 3x+2y \ge 0 \end{cases}$$

Let D be the region which represents the solution of the above system of inequalities. If (x, y) is a point lying in D , then the least value of $5x+6y+234$ is

A. 45 .

B. 150 .

C. 178 .

D. 423 .

(DSE-2018 PII Q34)

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5. The real part of $\dfrac{2i^{12}+3i^{13}+4i^{14}+5i^{15}+6i^{16}}{1-i}$ is

A. -3 .

B. -1 .

C. 1 .

D. 3 .

(DSE-2018 PII Q37)

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6. For $0^\circ \leq x < 360^\circ$, how many roots does the equation $6\cos^2 x = \cos x + 5$ have?

A. 2

B. 3

C. 4

D. 5

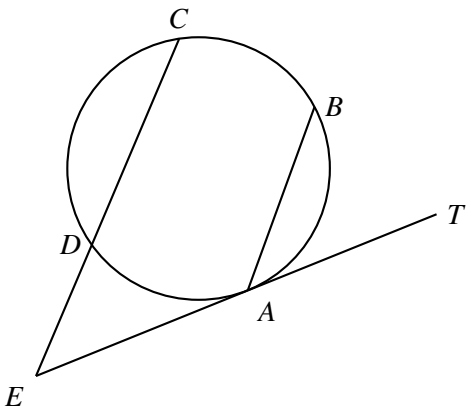
(DSE-2018 PII Q38)

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7. In the figure, TA is the tangent to the circle $ABCD$ at the point A . CD produced and TA produced meet at the point E . It is given that $AB = CD$, $\angle BAT = 24^\circ$ and $\angle AED = 72^\circ$. Find $\angle ABC$.



A. 60°

B. 66°

C. 72°

D. 78°

(DSE-2018 PII Q39)

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8. It is given that a is a positive constant. The straight line $2x+5y=a$ cuts the x-axis and the y-axis at the points P and Q respectively. Let R be a point lying on the y-axis such that the x-coordinate of the orthocentre of $\triangle PQR$ is 10 . Find the y-coordinate of R .

A. -25

B. -4

C. 4

D. 25

(DSE-2018 PII Q40)

I can finish _____ questions

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